

Vehicle Dynamics Engineer

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SUMMARY:

Master's student in Advanced Automotive Engineering at MUNER, specialising in Racing Car Design. Experienced in race engineering, simulation, and data-driven performance analysis through experience in Formula Student, Indian F4, and at Applus+ IDIADA. Skilled in Python, MBD simulation, tyre and vehicle performance analysis, with a strong ability to translate complex data into actionable engineering decisions in collaborative, high-pressure environments.

WORK EXPERIENCE:

Applus+ IDIADA

Vehicle Dynamics (MBD) Engineer | October 2023 – July 2024

- Performed Ride, Handling, K&C and other Vehicle Dynamics analyses in MSC ADAMS Car.
- Created full vehicle and component models in ADAMS.
- Used Python and ADAMS command line to implement an automation tool on ADAMS Car and reduce simulation times by 95%.
- Used FTire and CDTire models to conduct tyre simulations.

Indian Formula 4 Championship

Race Engineer | November 2023 – December 2023

- Participated in 4 weekends of the FIA Indian Formula 4 Championship as a Race Engineer.
- Worked with MP Motorsport engineers and mechanics to conduct tyre strategy and vehicle performance analyses.
- Participated and conducted driver data analysis, created competitor analysis tools using Motec i2 Pro and Wintax4.

Orion Racing India

Aerodynamics and Vehicle Dynamics Lead | September 2019 – July 2023

- Led a team of 20 students in the Aerodynamics and Vehicle Dynamics departments to design, build and test the university Formula Student car.
- Built a full carbon fiber aerodynamics package and initiated development of a custom tyre model on IPG Carmaker.

RELEVANT PROJECTS:

MUNER – Dallara Academy

Performance Analysis of single-seater race car on 7 Post Rig | September 2025 – January 2026

- Used data of a Dallara racecar from the 7 post rig to characterise vertical behaviour and damper settings.
- Created custom KPIs using MATLAB to conduct the analysis.

Applus+ IDIADA

Automation Toolkit | November 2023 – July 2024

- Created an Automation suite within MSC ADAMS Car for a client.
- The tool was used to load simulation settings, conduct simulations, process output data and generate automated reports using Python and Command Line.

EDUCATION:

2024 – 2027 | MotorValley University of Emilia-Romagna (University of Modena and Reggio Emilia)

Masters of Science – Advanced Automotive Engineering – Racing Car Design

2019 – 2023 | K.J Somaiya College of Engineering (University of Mumbai)

Bachelor of Technology – Mechanical Engineering

SKILLS:

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|----------------|----------------|------------|-------------------------|
| • Python | • IPG Carmaker | • MS Excel | • Trackside Engineering |
| • MSC Adams | • GitHub | • MS Word | • Data Analysis |
| • Motec i2 Pro | • Solidworks | • MATLAB | • Learning on the fly |
| • WinTax4 | • 3DEXperience | • Simulink | • Team player |

References available upon request